

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 01)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
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6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

01

Table 1 shows the number of adult great crested newts caught in bottle traps in a pond over five years, and the number of ponds in the area in which the species was detected using environmental DNA (eDNA).

Table 1

Year	Adults caught in traps	Ponds with newts detected by eDNA (out of 20)
2019	14	6
2020	22	9
2021	18	11
2022	31	13
2023	27	15

01 . 1

Calculate the percentage increase in the number of ponds where newts were detected between 2019 and 2023.

[2 marks]

Answer %

01 . 2

Explain how eDNA detects the presence of newts without catching them.

[2 marks]

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01 . 3

Suggest why trap catches are a less reliable measure of population change than the eDNA survey.

[2 marks]

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Evaluate the contribution of new technologies to ecological monitoring and wildlife conservation.

ESP/7447/2/SET01

Extra space

15

0 2 . 1

Explain how GPS mapping, drone surveys and computer-based yield monitoring can reduce the environmental impacts of agriculture.

[3 marks]

0 2 . 2

Explain why productive agricultural technologies are less available in LEDCs.

[2 marks]

0 2 . 3

Explain how buying fairtrade food affects producers.

[1 mark]

0 2 . 4

Explain how nitrate from farmland can threaten human health.

[2 marks]

0 2 . 5

Give **two** agricultural sources of methane.

[2 marks]

1

2

10

0 3

Table 1 shows some medicines that were discovered in wild species.

Table 1

Medicine	Source species	Use
Taxol	Yew tree	Cancers of the breast, ovary, lung, bladder and prostate
Quinine	Cinchona tree	Malaria
AZT	Caribbean marine sponge	HIV / AIDS
Diosgenin (steroids)	Mexican yam	Contraceptive pill, cortisone

0 3 . 1

Explain why many plants produce chemicals that turn out to be useful medicines.

[2 marks]

03 . 2

Explain why the medicines in Table 1 provide an argument for conserving whole habitats rather than individual species.

[2 marks]

03 . 3

Describe how **one** named species is being studied to help solve a human health problem other than by providing a medicine.

[2 marks]

03 . 4

Discuss the extent to which the resources and ecosystem services provided by other species justify the conservation of biodiversity.

[9 marks]

Do not write outside the box

Extra space

15

04 . 1

Explain the meaning of each of the following terms and explain why such species are given conservation priority.

[8 marks]

EDGE species

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Endemic species

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Keystone species

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Flagship species

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04 . 2

Explain why the degree of population dispersal is considered when setting priorities.

[2 marks]

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10

0 5 . 1

Describe **two** case studies of biological control, one successful and one that caused problems.

[4 marks]

0 5 . 2

Explain how pheromone traps can be used in pest control.

[2 marks]

0 5 . 3

Explain why the use of antibiotics as growth promoters in livestock is a concern for human health.

[3 marks]

0 5 . 4

Give **one** way farmers can support crop pollinators.

[1 mark]

10

0 6 . 1

Which of the following features of a plantation would most increase its wildlife value? Tick one box.

[1 mark]

A Close planting of a single non-indigenous species

☐

B Mixed indigenous species with a mixed age structure

☐

C Clear-felling on a 40-year rotation

☐

D Removal of dead wood

☐

0 6 . 2

Describe **four** features of sustainably managed forests and explain how each benefits biodiversity or productivity.

[4 marks]

1

2

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4

5

0 7 . 1

Give **two** natural processes that are driven by solar energy.**[2 marks]**

1

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2

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0 7 . 2

Explain why natural processes can take place at low temperatures.

[2 marks]

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0 7 . 3

Give **one** human activity that uses high temperatures to achieve a result that natural systems achieve at low temperature.**[1 mark]**

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5

0 8 . 1

Which feature of the Earth prevents most atmospheric gases from escaping into space? Tick **one** box.**[1 mark]**

A Its mass

☐

B Its magnetic field

☐

C Its distance from the Sun

☐

D Its speed of rotation

☐

0 8 . 2

Explain how the speed of rotation of the Earth and the tilt of its axis affect the conditions for life.

[2 marks]

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08 . 3

Explain why the presence of liquid water on Earth depends on **both** the distance from the Sun **and** atmospheric pressure.

[2 marks]

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5

09 . 1

State what is meant by **hybrid vigour**.

[1 mark]

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09 . 2

Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

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09 . 3

Give **two** potential applications of cloning to livestock farming.

[2 marks]

1

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2

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5

1 0 . 1

Explain how the relationship between coral polyps and zooxanthellae allows a high biomass of life in nutrient-poor tropical seas.

[4 marks]

1	0	.	2
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Explain why growing coral reefs help to resist climate change.

[2 marks]

1	0	.	3
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Discuss the importance of tropical coral reefs and evaluate the methods used to conserve them.

[9 marks]

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Extra space

Do not write outside the box

15

Do not write outside the box

Write an essay on one of the following topics.

1	1	.	1
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Agriculture involves the control of abiotic and biotic factors to maximise food production. Discuss how this control can be achieved and evaluate its environmental impacts.

[25 marks]

OR

1 1 . 2

Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

[illegible]

Do not write outside the box

Handwriting practice area with 25 horizontal dotted lines.

END OF QUESTIONS